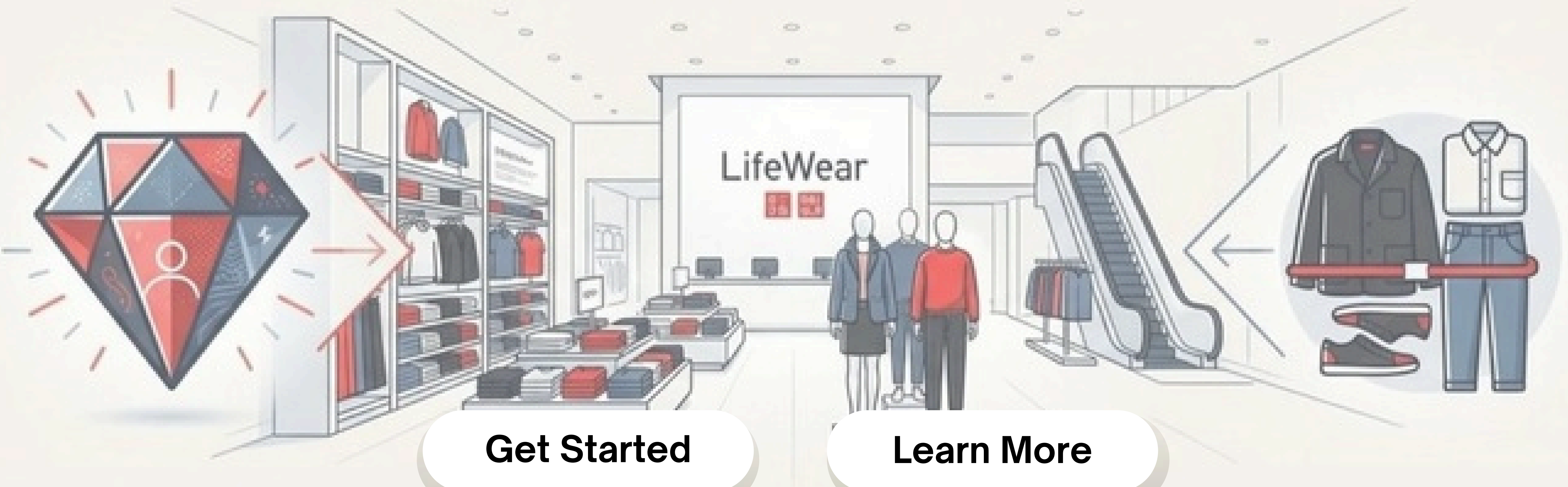
Group 1

SELF-SUPERVISED LEARNING IN RECOMMENDER SYSTEMS



Get Started

Learn More

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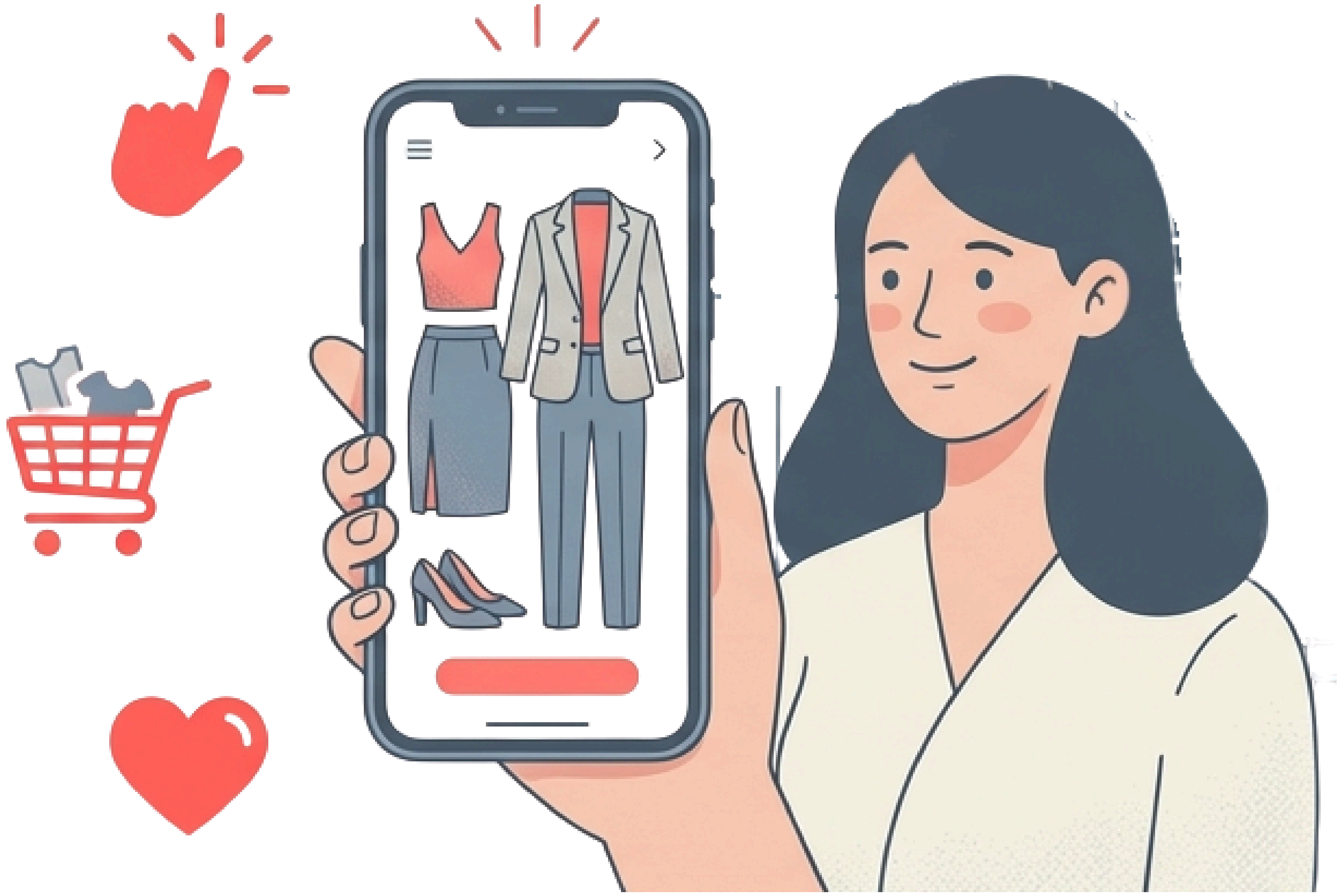
- CroRec application
- Predicting with RL & Improving Diversity with AL



RECOMMENDER SYSTEMS

Definition

Recommendation systems are used to solve the problem of "information overload" by helping each person pick out things they might like.

- A predict function
  - $f(\mathbf{A}, \mathbf{X}, \mathbf{u}, \mathbf{v}) \rightarrow \mathbf{y_{u,v}}$
- Two sets
  - User: $\mathbf{U} = \{u_1, u_2, \ldots\}$
  - Items: $\mathbf{V} = \{v_1, v_2, \ldots\}$
- One interaction matrix $\mathbf{A_{ij}}$
  - $A_{ij} = 1 \rightarrow$ user i and item j interact
  - $A_{ij} = 0 \rightarrow$ no interaction
- distinct auxiliary data **X**
  - knowledge graphs and a user relationships

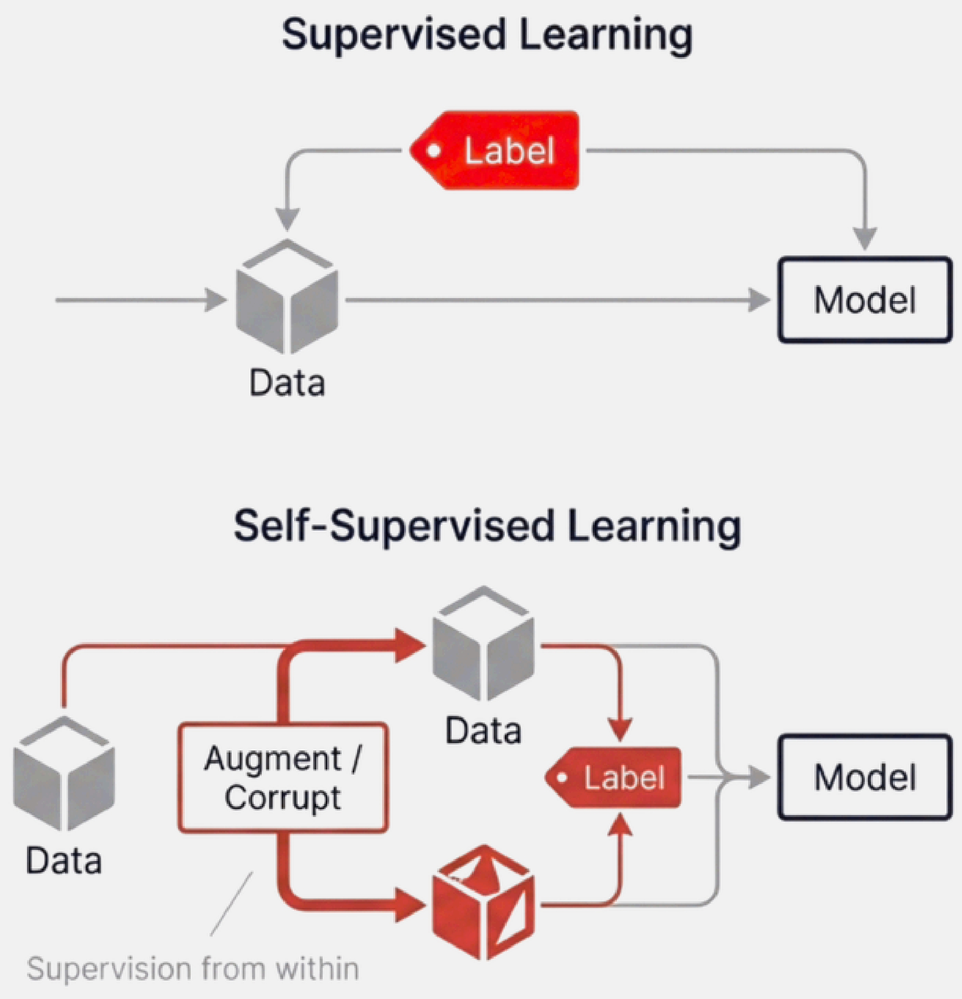
Motivates

- **Data sparsity** in reality.
- Supervised learning is powerful, but requires many labels.
- A common recommendation scenario is that user behavior is sparse, and many items remain untouched

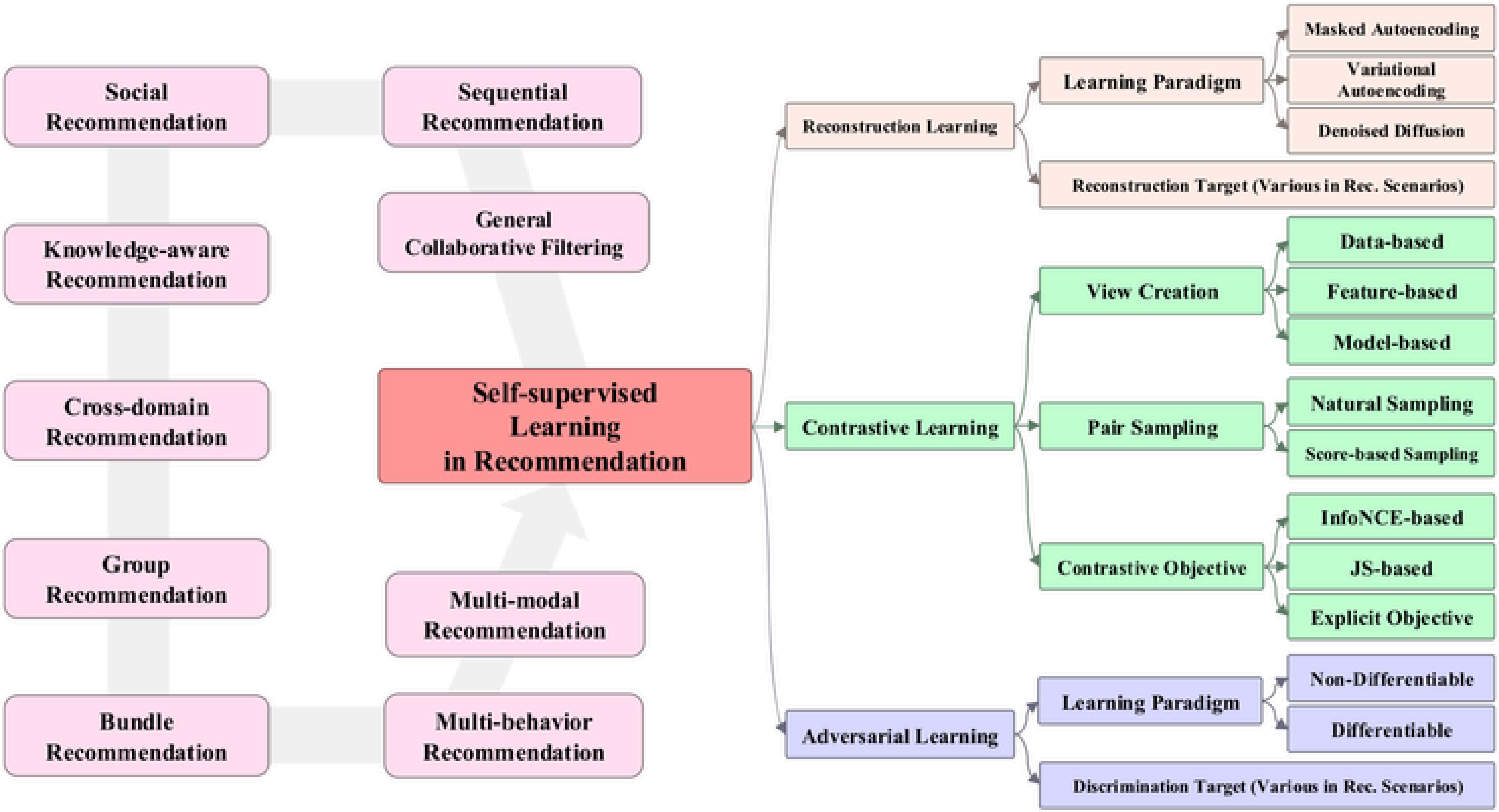
1. Collected 170+ papers
2. Created an open-source library SSLRec: Contains datasets, training and testing procedures, and many recommendation models.)
3. Proposed a new taxonomy (3 categories: Contrastive / Reconstructive / Adversarial.)

Self-Supervised learning

- Self-supervised learning → Generating training tasks using materials



STRUCTURE


Social Recommendation
Sequential Recommendation
General Collaborative Filtering
Knowledge-aware Recommendation
Cross-domain Recommendation
Self-supervised Learning in Recommendation
Group Recommendation
Multi-modal Recommendation
Bundle Recommendation
Multi-behavior Recommendation
Reconstruction Learning
Learning Paradigm
Masked Autoencoding
Variational Autoencoding
Denoised Diffusion
Reconstruction Target (Various in Rec. Scenarios)
Contrastive Learning
View Creation
Data-based
Feature-based
Model-based
Pair Sampling
Natural Sampling
Score-based Sampling
Contrastive Objective
InfoNCE-based
JS-based
Explicit Objective
Adversarial Learning
Learning Paradigm
Non-Differentiable
Differentiable
Discrimination Target (Various in Rec. Scenarios)

STRUCTURE

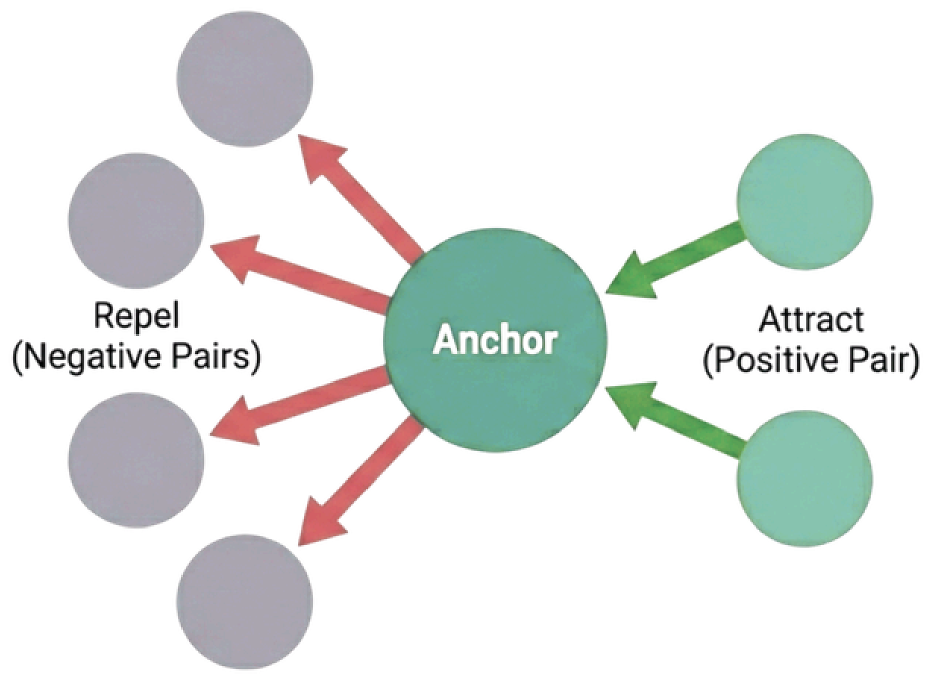
Taxonomy of SSL

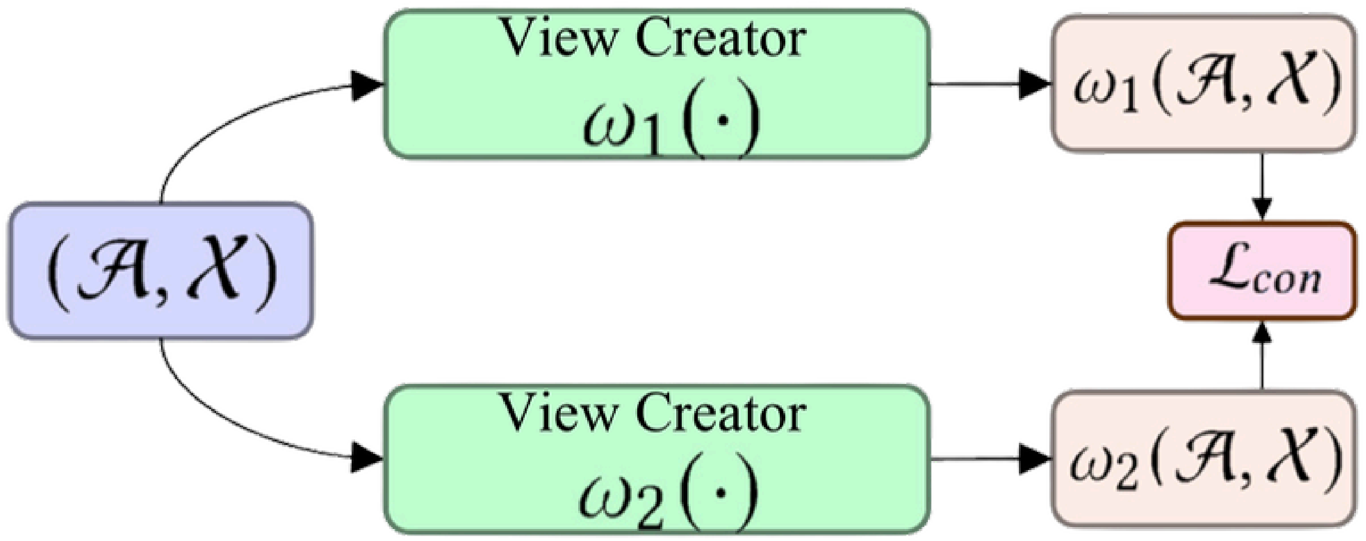
- Contrastive Learning (CL)
- Reconstructive Learning (RL)
- Adversarial Learning (AL)

Application

General Collaborative Filtering	Sequential Recommendation	Social Recommendation
Knowledge-Aware Recommendation	Cross-domain Recommendation	Bundle Recommendation
Group Recommendation	Multi-Behavior Recommendation	Multi-Modal Recommendation

CONTRASTIVE LEARNING(CL)

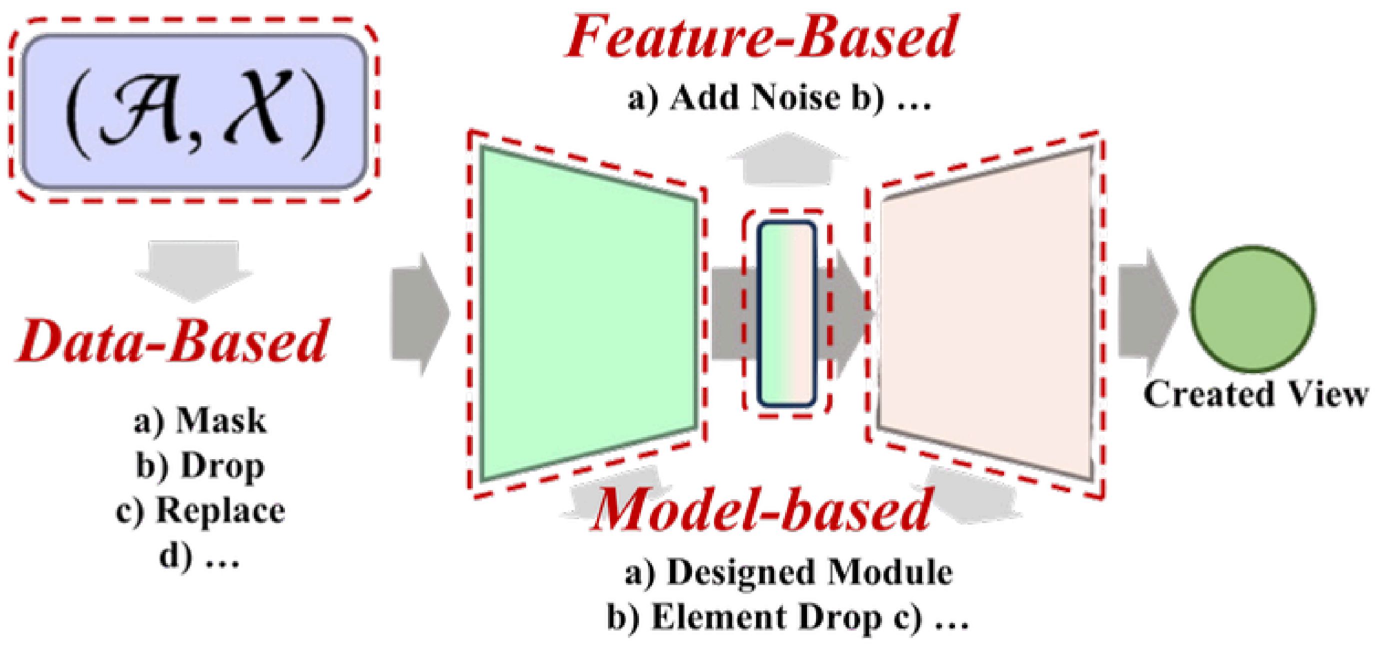




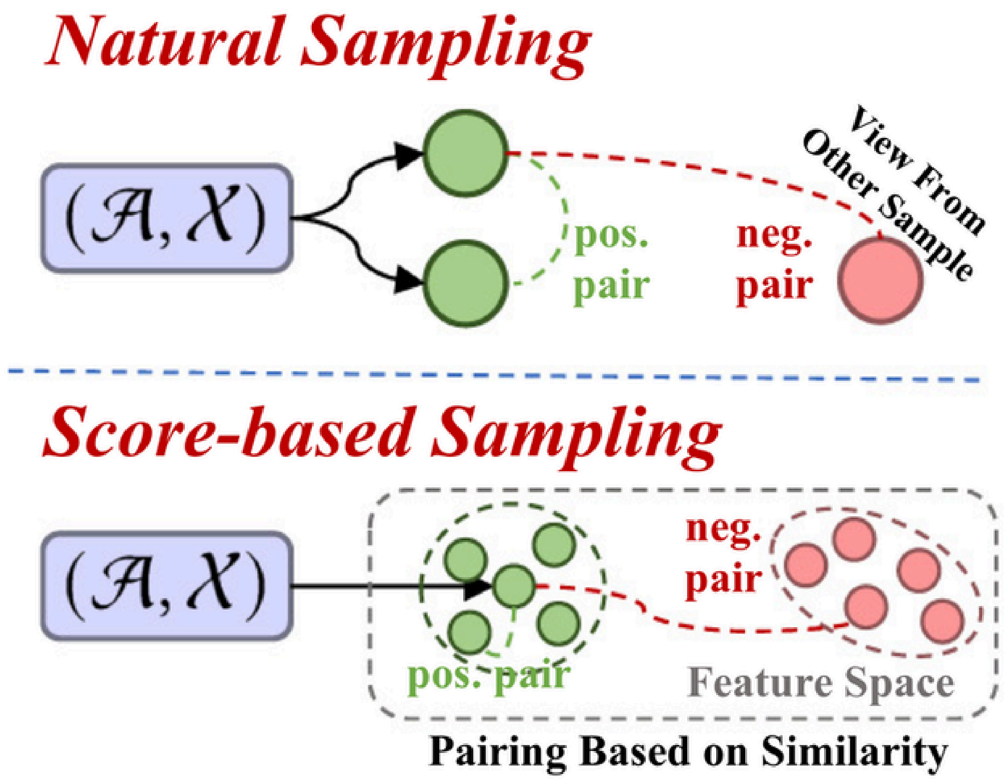
Contrastive learning teaches a model to understand what something is by learning what it's similar to and what it's different from. The primary objective is to maximize the agreement augmented from the same data point.

Perform two different operations on the data: $\omega_1(A, X)$ and $\omega_2(A, X)$. For example: Randomly drop some edges/nodes in the graph. Add some noise to the embedding. Feed each into encoders E_1 and E_2. Finally, use a comparison loss L_con to make the two views as similar as possible.
Formula: min $L\_con(E_1 \circ \omega_1(A,X), E_2 \circ \omega_2(A,X))$

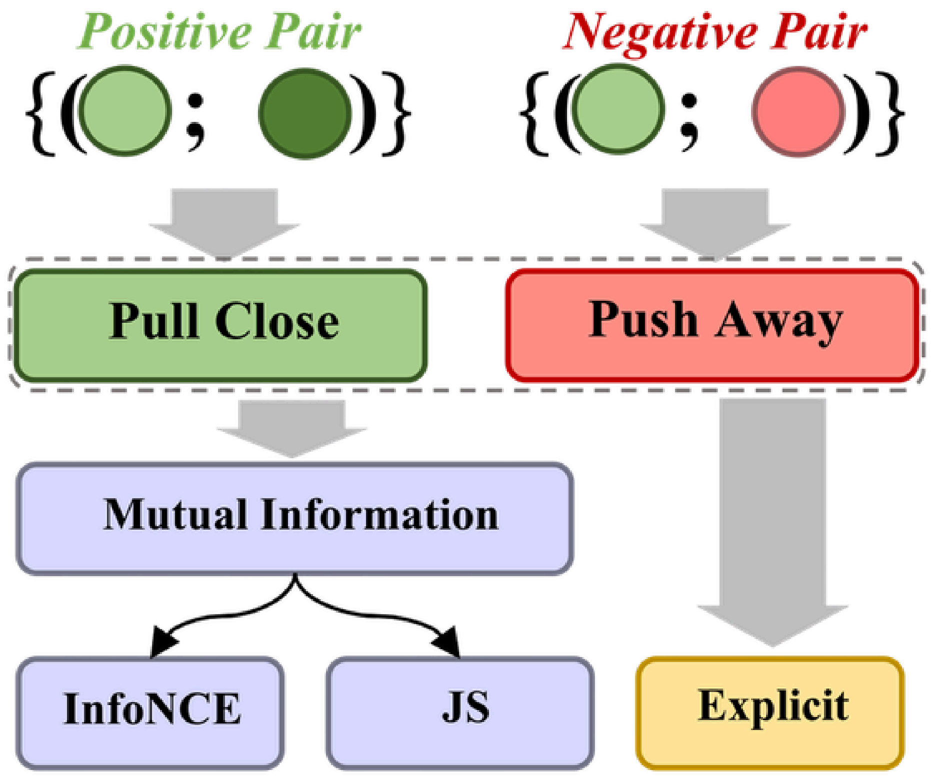
CONTRASTIVE LEARNING(CL)



1. View Creation



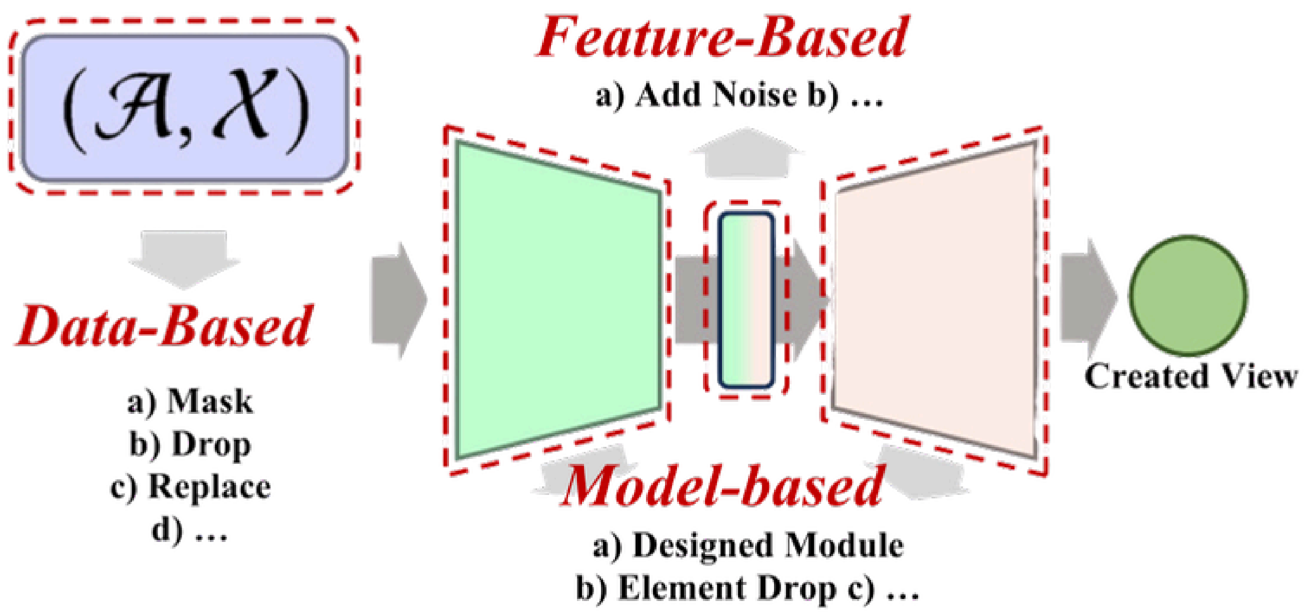
2. Pair Sampling



3. Contrastive Objective

CONTRASTIVE LEARNING(CL)

- VIEW CREATION

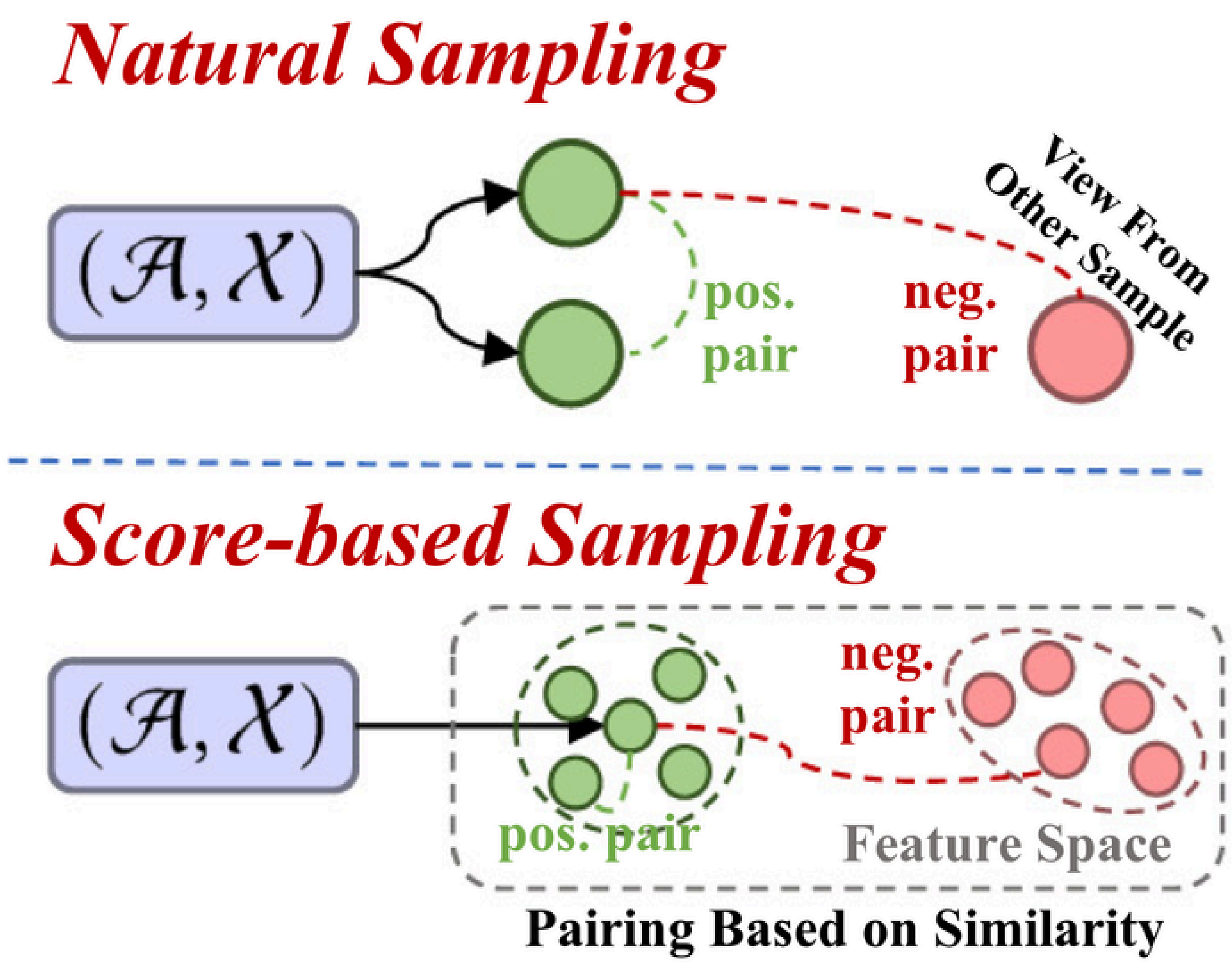


Generating different perspectives of the data

- Data-based: Augmenting input data (e.g., node/edge dropout on graphs, masking/cropping sequences).
- Feature-based: Adding noise or perturbations to hidden embeddings during model inference.
- Model-based: Using learnable neural modules (e.g., intent disentanglement) to generate adaptive views.

CONTRASTIVE LEARNING(CL)

- PAIR SAMPLING

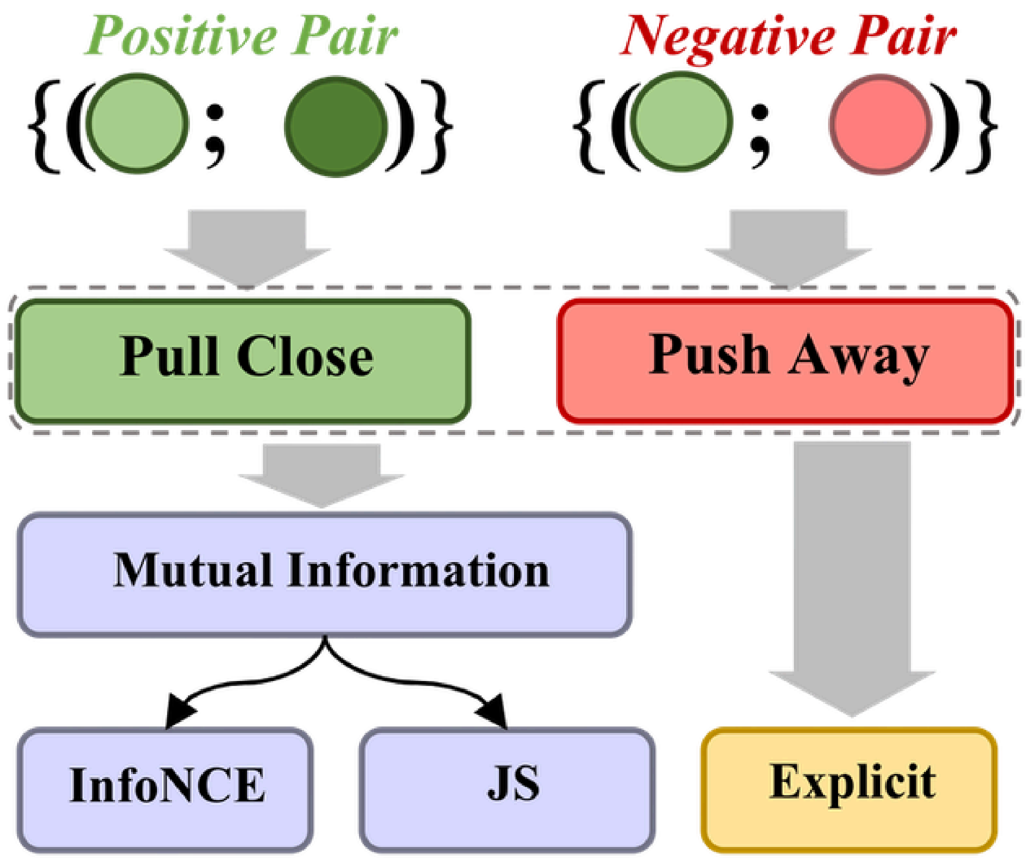


Defining what to compare

- Natural Sampling: Positive pairs are views from the same data sample. Negative pairs are views from different samples.
- Score-based Sampling: Using similarity scores or clustering to define positive/negative pairs.

CONTRASTIVE LEARNING(CL)

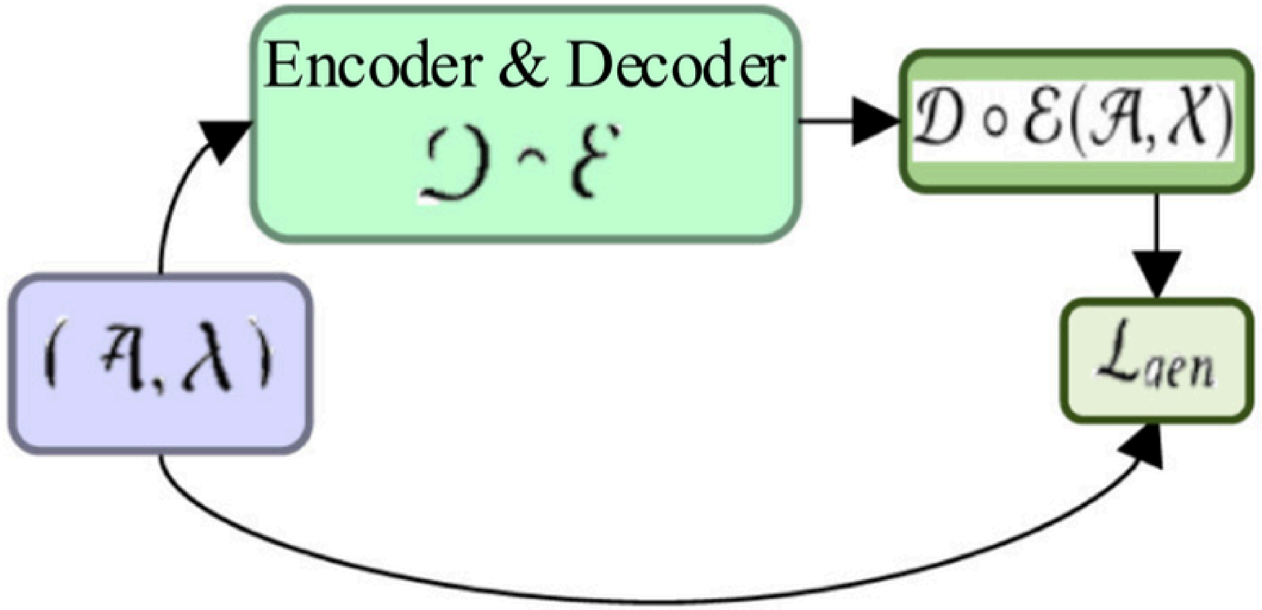
- CONTRASTIVE OBJECTIVE



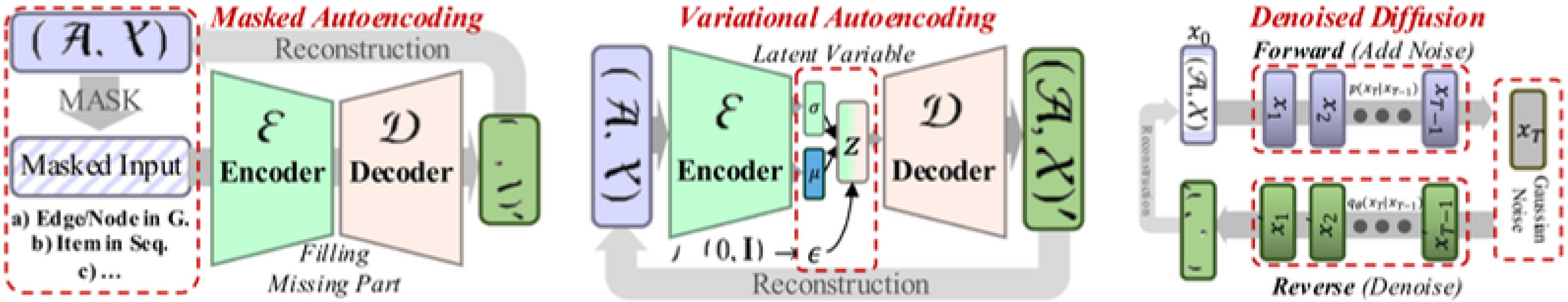
The loss function for optimization

- InfoNCE-based: The most common objective, a variant of Noise Contrastive Estimation.
- JS-based: An alternative lower bound estimation using Jensen–Shannon divergence.
- Explicit: Directly minimizing distance (e.g., MSE) or maximizing cosine similarity between positive pairs.

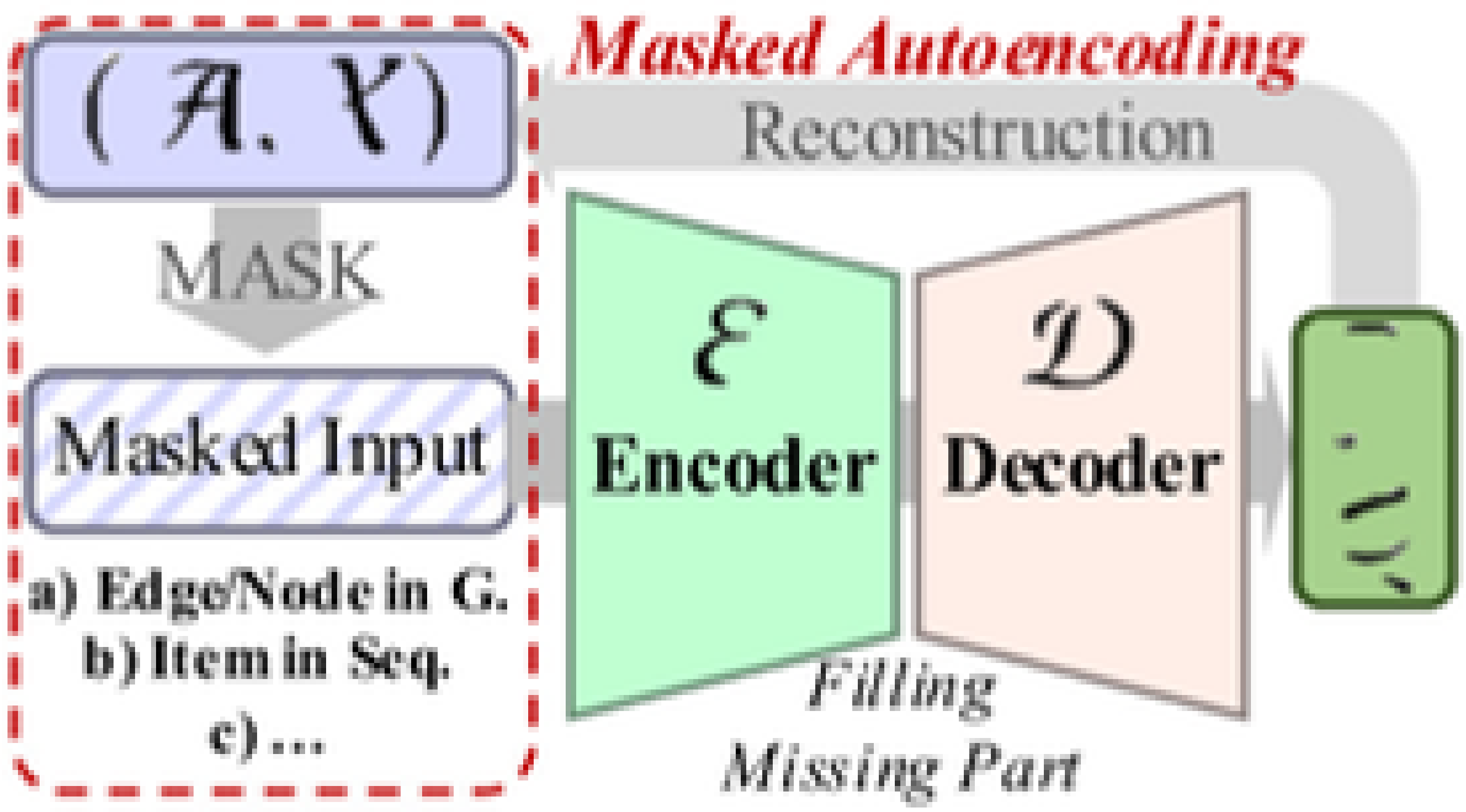
RECONSTRUCTIVE LEARNING (RL)



- Maximize the likelihood estimation of the real data distribution
- Capture the underlying patterns in the data

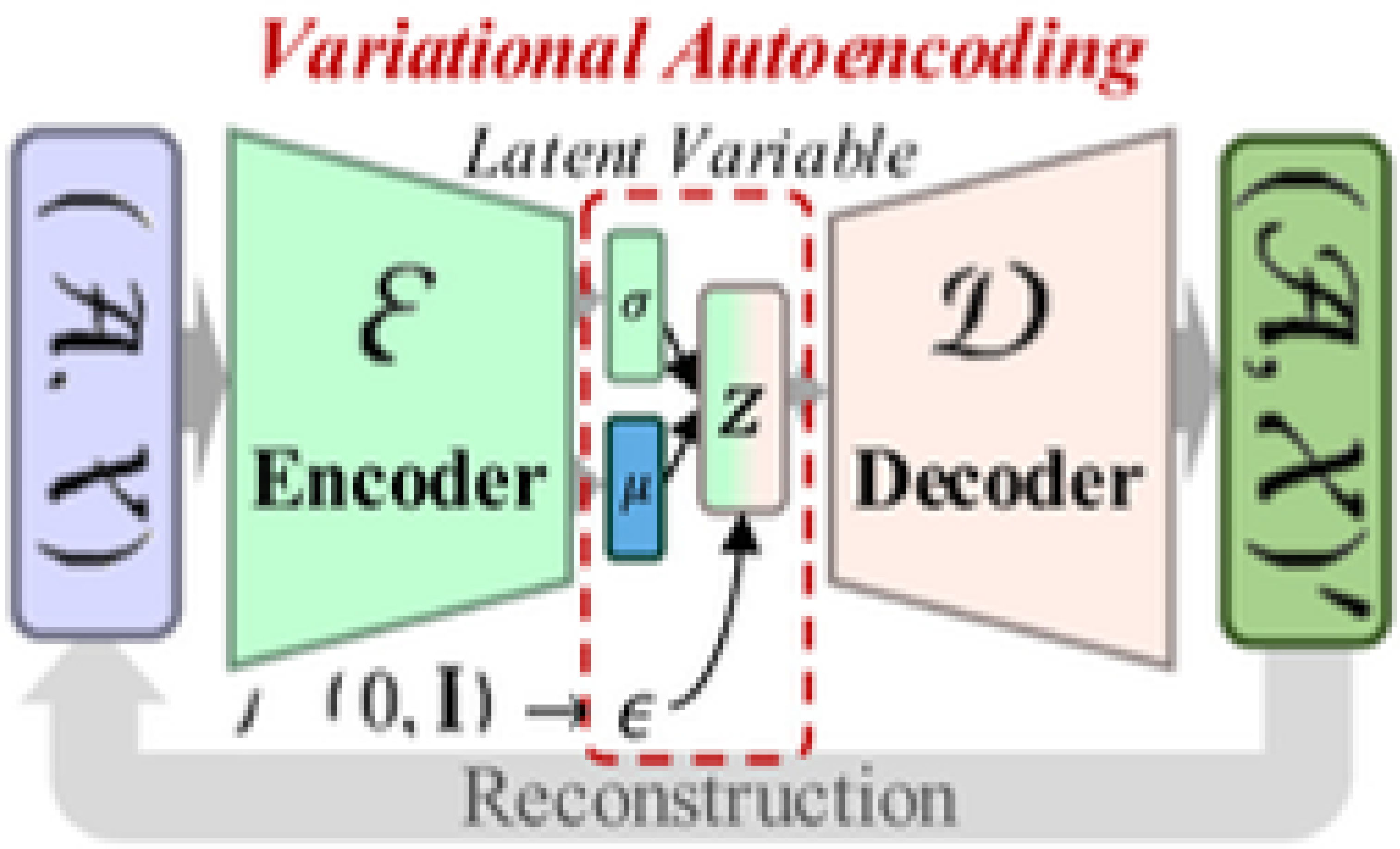


RECONSTRUCTIVE LEARNING (RL)——MASKED AUTOENCODING(MAE)



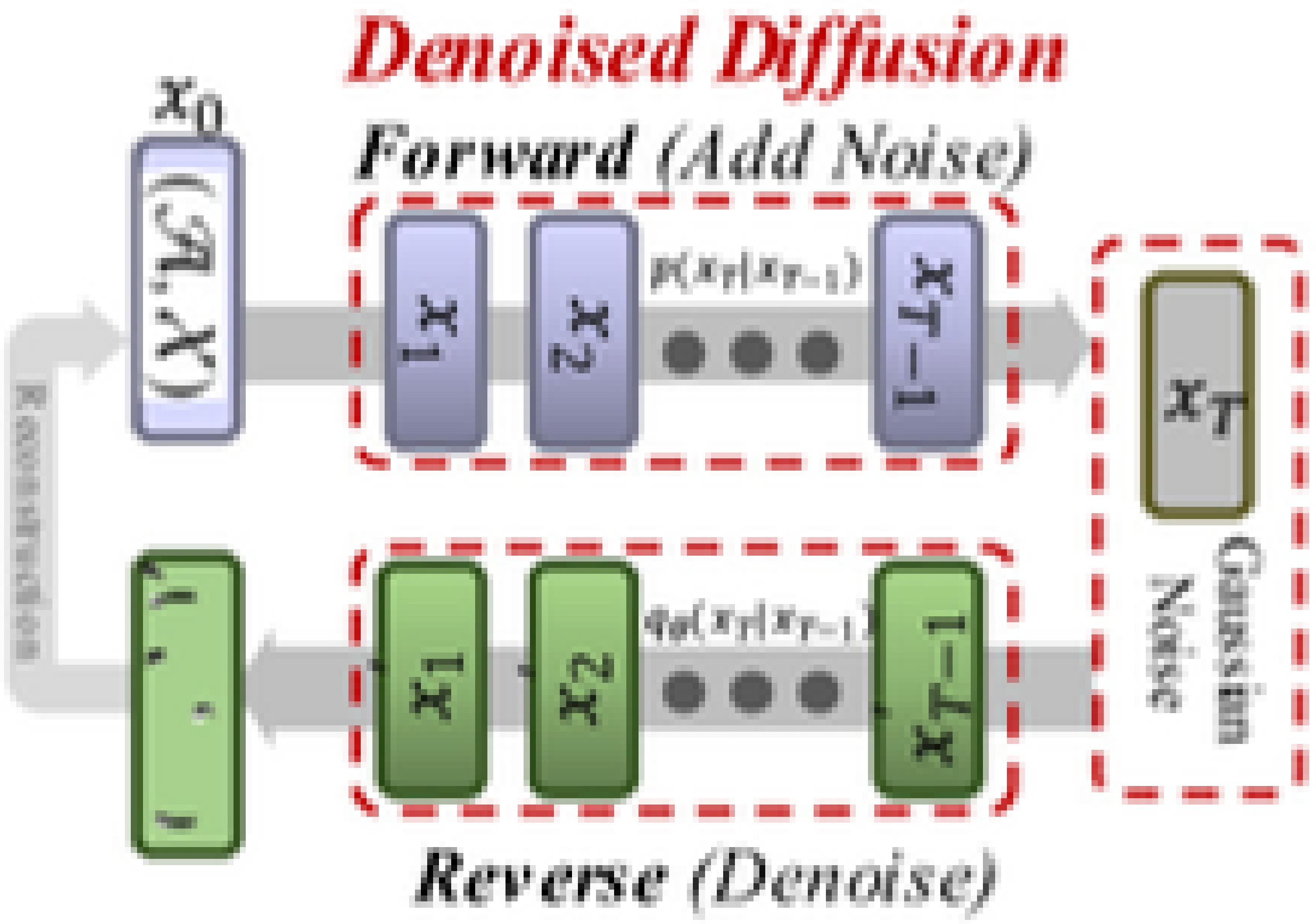
- Learning process follows a mask-reconstruction approach, mask some
- Data can be in **item sequences** or **user-item interaction** graphs
  - **item sequences** : randomly masked and fed into a transformer model
  - **graph data**: masking occurs at the edge or node level

RECONSTRUCTIVE LEARNING (RL)——VARIATIONAL AUTOENCODING



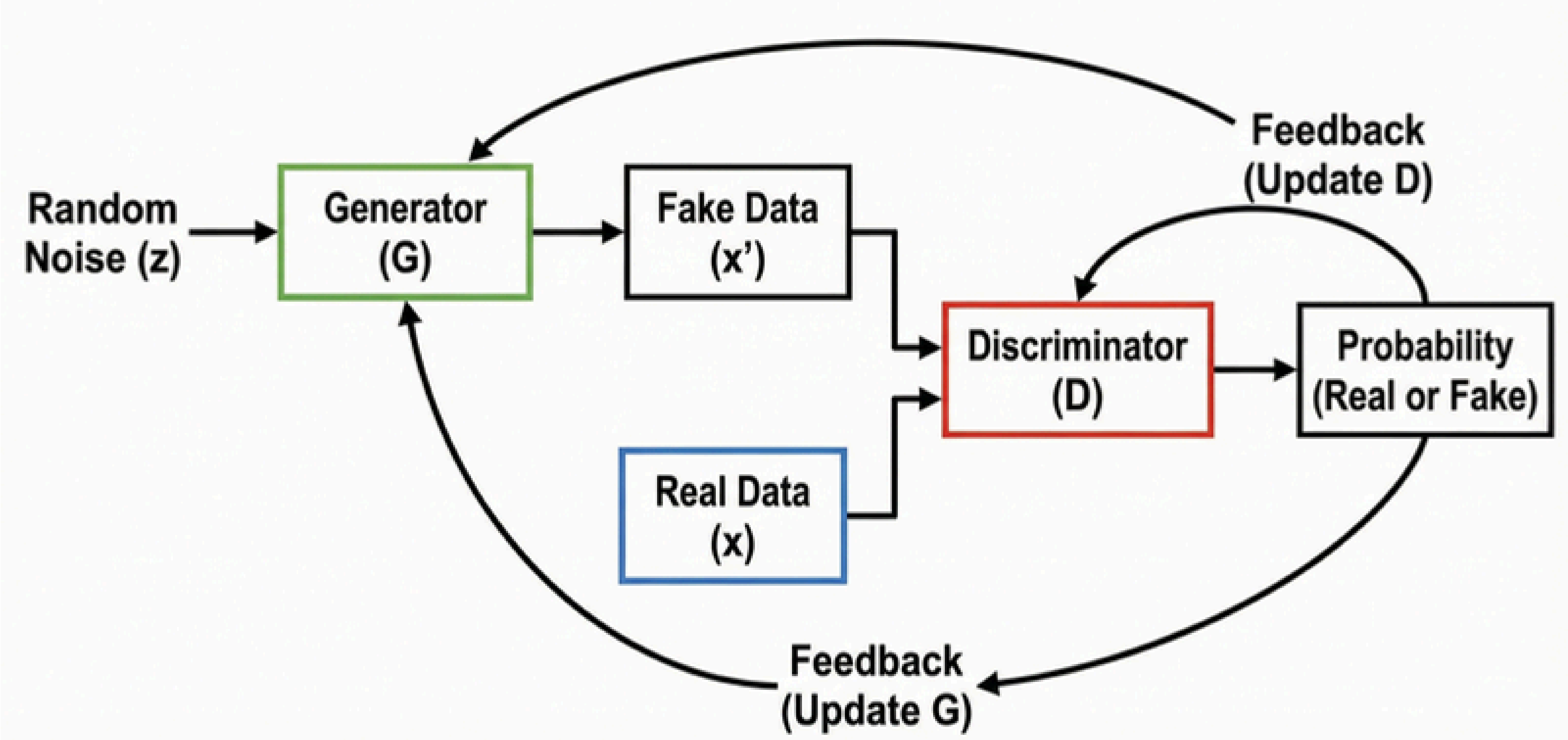
- Map the input data to latent factors that follow a normal Gaussian distribution and reconstruct the data from sampled latent variables.
- Use KL-divergence during training to enforce the latent space to match a Gaussian distribution.

RECONSTRUCTIVE LEARNING (RL)



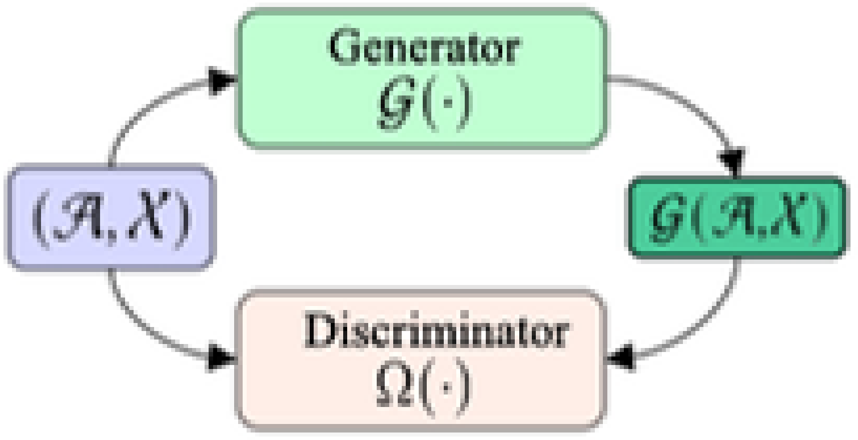
- Reconstruct the data samples by reversing a noising process
- In the forward process, Gaussian noise is added to the original data in multiple steps
- In the reverse process, the model learns to remove the noise from the noisy versions, recovering the original data step by step.
- Trained to denoise the data at each step, learning to capture minor changes in the complex generation process.

ADVERSARIAL LEARNING (AL)

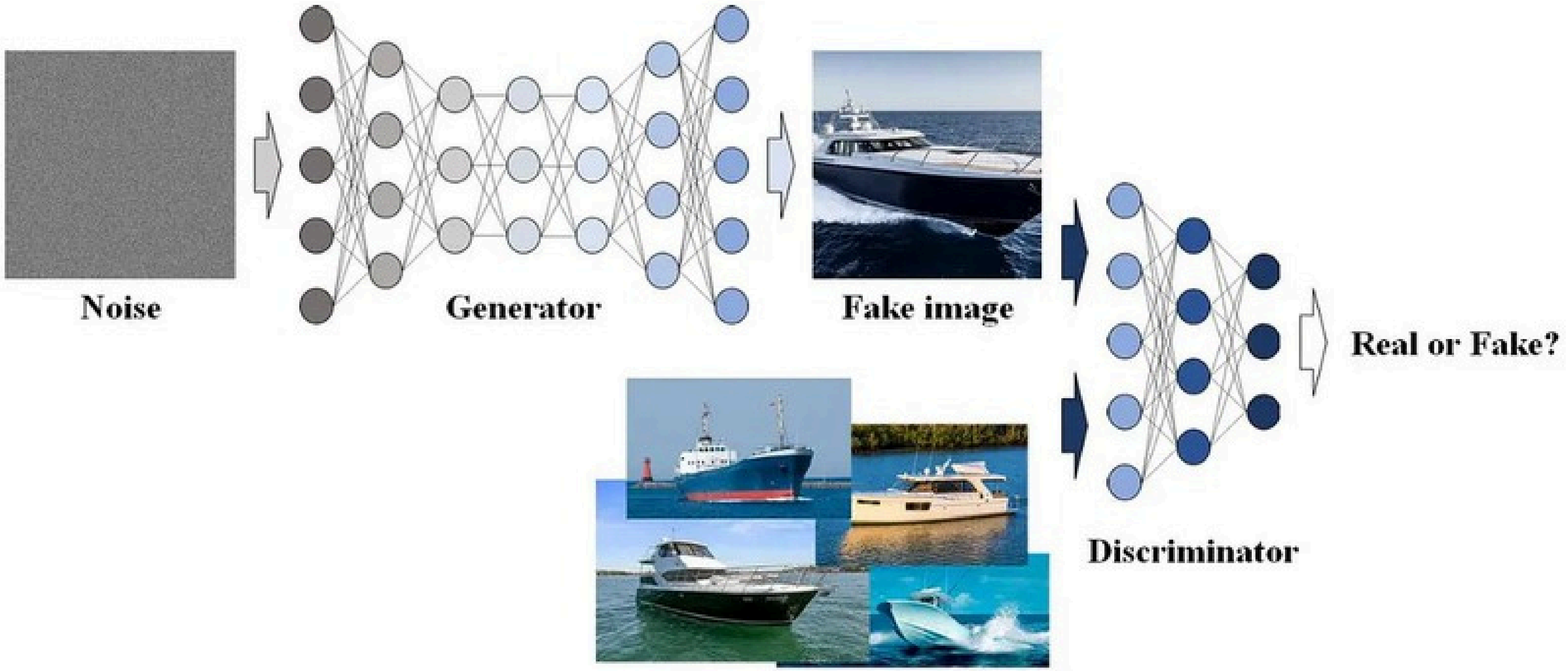


Graph Generated by Gemini

ADVERSARIAL LEARNING (AL)

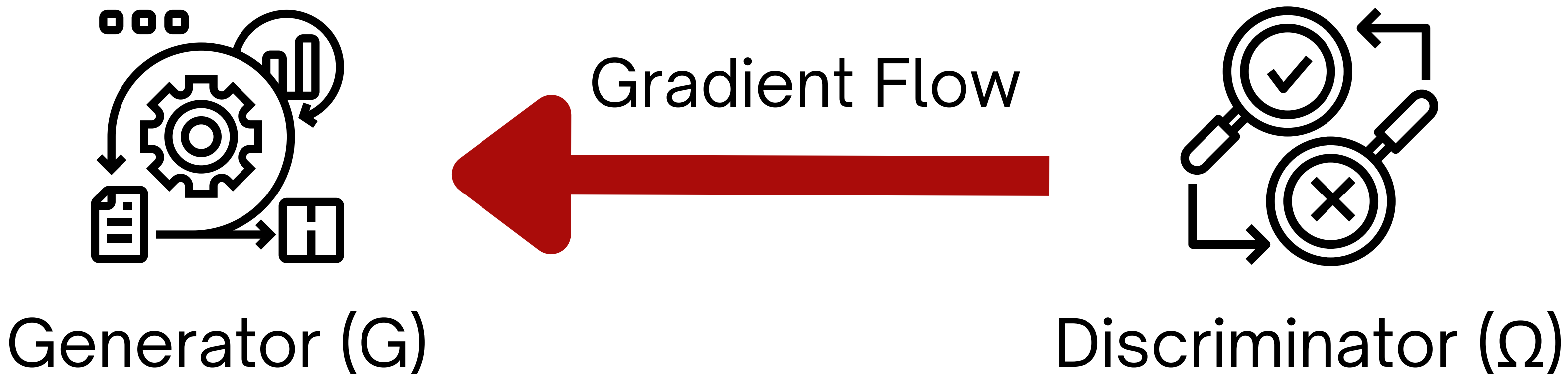


(a) Adversarial Learning



ADVERSARIAL LEARNING

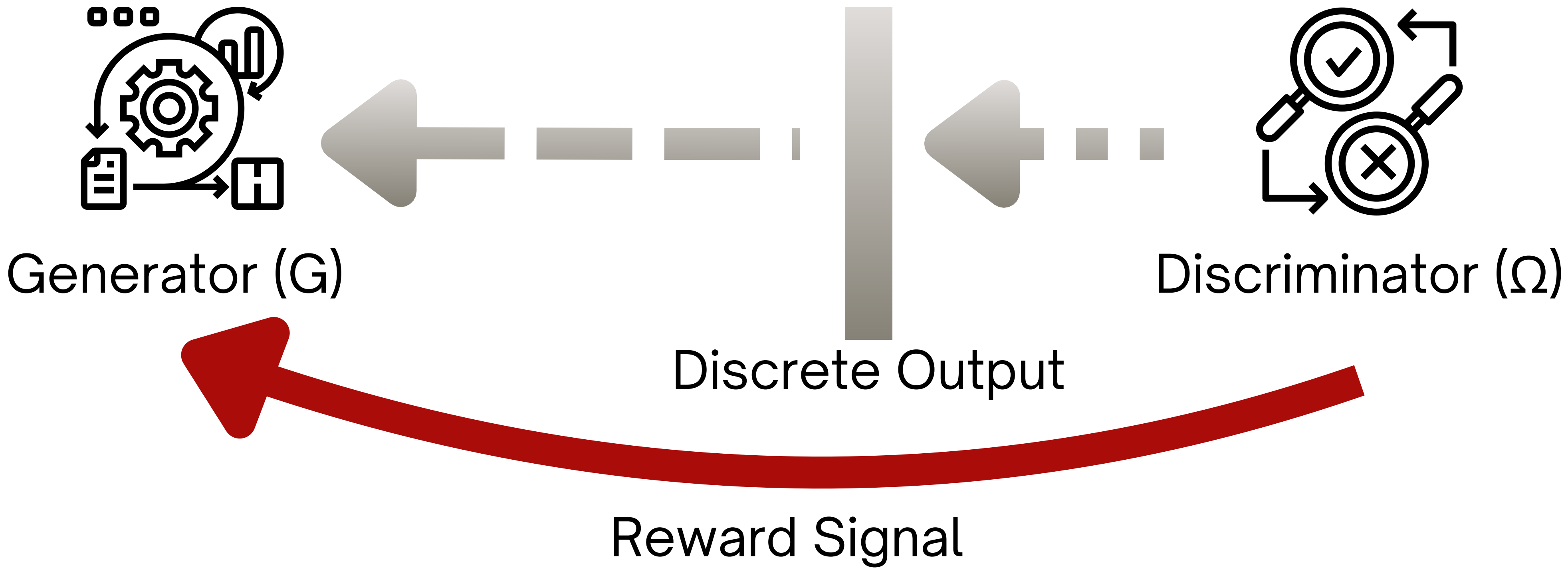
DifferentiableAL



Used for continuous data (e.g., user feature vectors).

Applicable when objects are represented in a continuous space (e.g., latent representations or features), allowing the discriminator's gradient to backpropagate naturally to the generator.

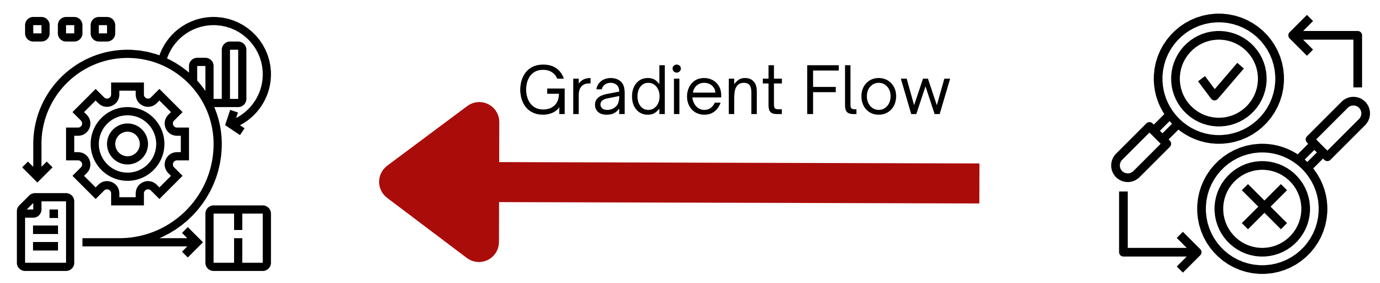
Non-differentiableAL



Used for discrete outputs (e.g., recommended item indices). Since gradients cannot flow directly, optimization typically involves reinforcement learning using policy gradients (e.g., IRGAN).

ADVERSARIAL LEARNING

DifferentiableAL



Generator (G) Discriminator (Ω)

Used for continuous data (e.g., user feature vectors).

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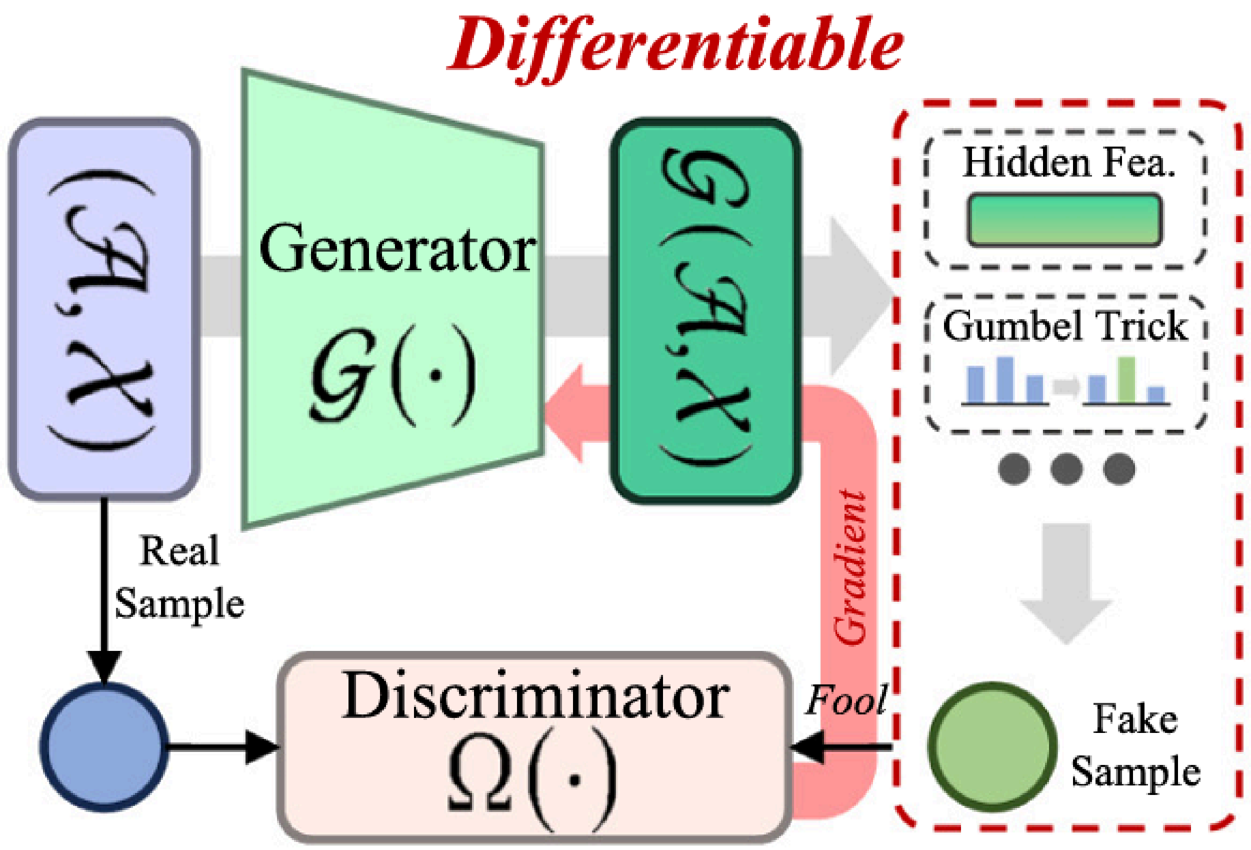
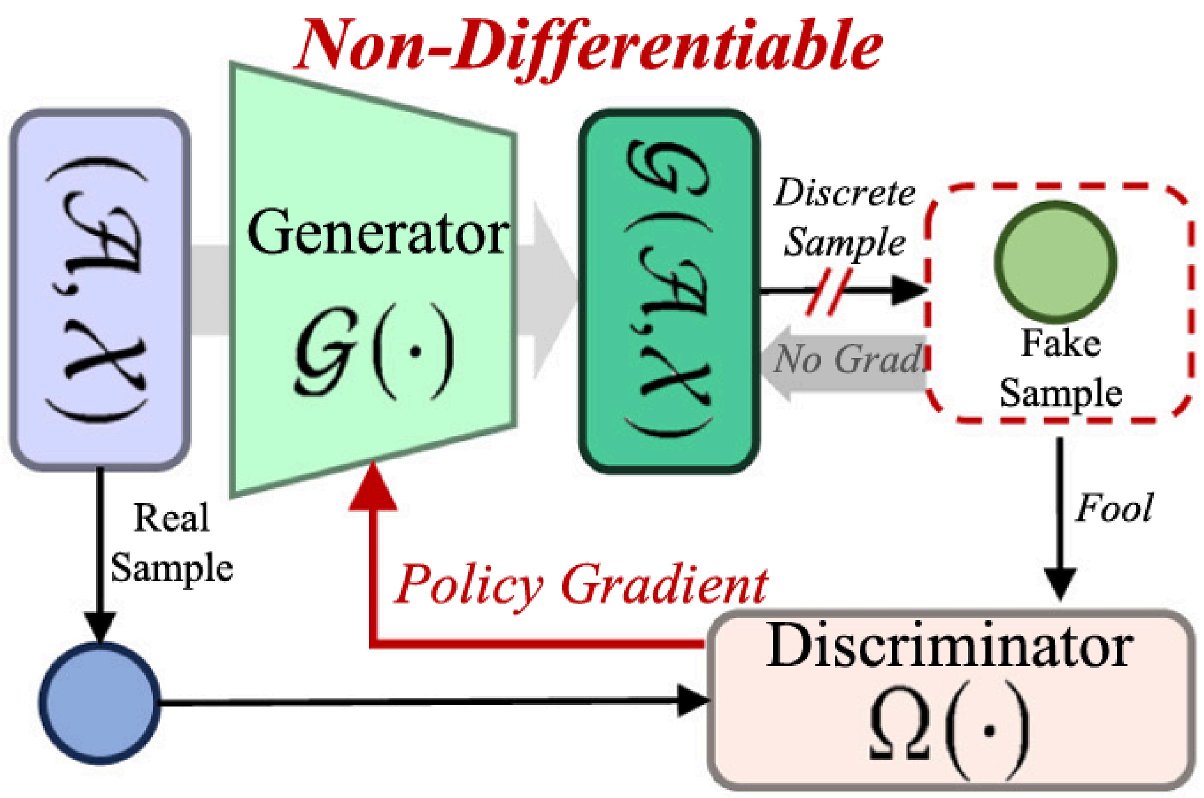
Non-differentiableAL

Generator ... inator (Ω)



Use... item indices). Since gradients cannot flow directly, optimization typically involves reinforcement learning using policy gradients (e.g., IRGAN).

ADVERSARIAL LEARNING



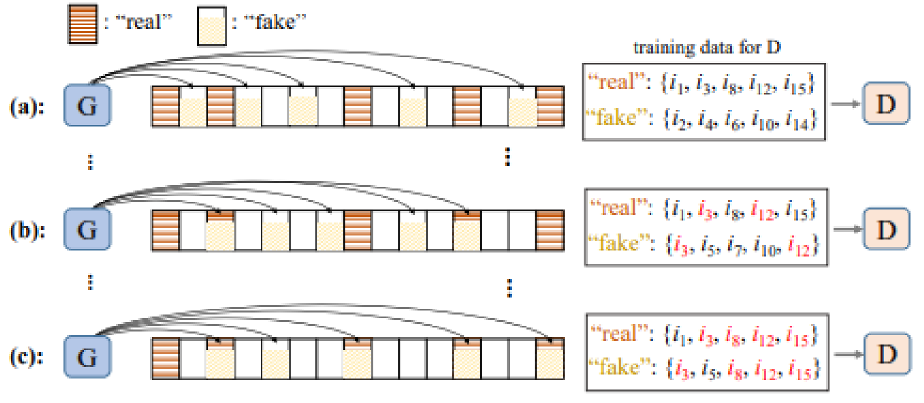
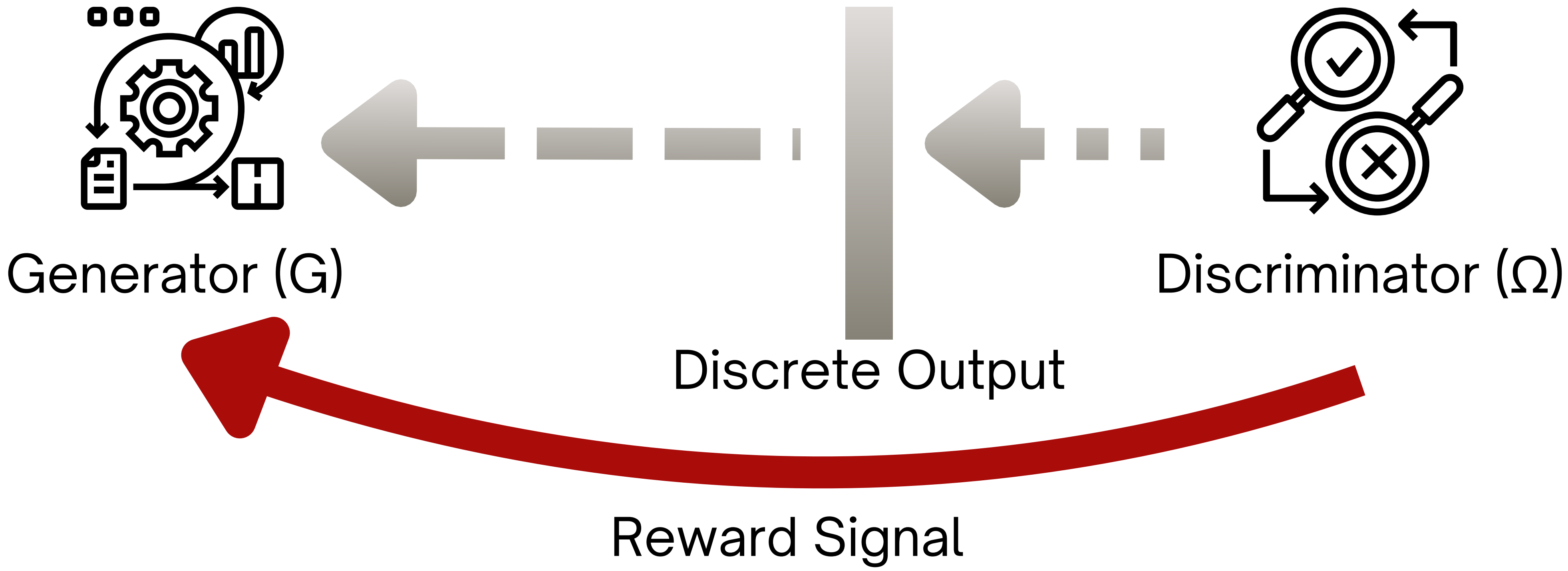


Figure 1: Discrete item index generation.

Discrete Output

Non-differentiableAL



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STRUCTURE

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- Reconstructive Learning (RL)
- Adversarial Learning (AL)

Application

General Collaborative Filtering	Sequential Recommendation	Social Recommendation
Knowledge-Aware Recommendation	Cross-domain Recommendation	Bundle Recommendation
Group Recommendation	Multi-Behavior Recommendation	Multi-Modal Recommendation

CL, RL, AL APPLICATION

General Collaborative Filtering, CF	Sequential Recommendation, SeqRec	Social Recommendation, SocRec
CL Create diverse views, sample +/- pairs, and align representations.	**CL** Build varied sequence views and aligns them.	**CL** Generate multiple social-aware views, construct +/- pairs, and align them through contrastive objectives to strengthen user–user and user–item relational representations in social recommendation.
RL Leverage generative learning techniques to reconstruct user-item interactions, generating self-supervised signals to enhance performance	**RL** Generate sequence item data or user-item graphs to provide auxiliary signals, enhance the recommendation process.	
AL Improves recommendation accuracy by optimizing outputs via a discriminator, enhancing robustness against noise.	**AL** Train discriminators to distinguish real/generated sequences, optimizing the generator to produce realistic item sequences.	**AL** Refining the quality of user interactions by extracting clean and representative social relations in social data and enhancing model robustness

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CL, RL, AL APPLICATION

Knowledge-Aware Recommendation, KnoRec	Cross-domain Recommendation, CroRec	Bundle Recommendation, BunRec
CL Build knowledge-enhanced views from KG and user–item graphs, forms meaningful pairs, and align them via InfoNCE to improve semantic recommendation representations.	**CL** Build multi-domain views and aligns them via contrastive objectives to transfer knowledge across domains	**CL** Construct multi-level user–item–bundle views, contrast dual representations, and align cross-granular features with InfoNCE to enhance bundle recommendation performance.
RL Discover valuable knowledge triplets through generative tasks while mitigating the negative impact caused by noisy or irrelevant knowledge.	**RL** Align and transfer knowledge between the source and target domains, facilitate knowledge sharing and adaptation	**RL** Adopt autoencoder paradigm to learn meaningful representations by masking and reconstructing critical structures to enhance model robustness
	AL Generate domain-invariant features that improve generalization across domains.	

CL, RL, AL APPLICATION

Knowledge-Aware Recommendation, KnoRec

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Build knowledge-enhanced views from KG and user–item graphs, forms meaningful pairs, and align them via InfoNCE to improve semantic recommendation representations.

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Discover valuable knowledge triplets through generative tasks while mitigating the negative impact caused by noisy or irrelevant knowledge.

Cross-domain Recommendation, CroRec

CL

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Bundle Recommendation, BunRec

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CL, RL, AL APPLICATION

Group Recommendation, GroRec

CL
GroRec CL constructs multi-granular user–group and group–item views by leveraging group structures, user interactions, and hypergraph-based augmentations. These diverse views are aligned through contrastive objectives, allowing the model to capture complex group dynamics and significantly enhance group-level recommendation performance.

Multi-Behavior Recommendation, MbeRec

CL
Create behavior-specific multi-view representations and align them to capture fine-grained user–item behavior patterns for stronger multi-behavior recommendation performance.

RL
Adopt the VAE paradigm to reconstruct multi-behavior interactions and generate informative auxiliary signals that enhance behavior-aware representation learning.

Multi-Modal Recommendation, MmoRec

CL
Build multi-modal views and align them for stronger fused item representations.

RL
Employ the graph VAE to transform modality data into a latent space, utilize generation tasks to guide the learning

AL
Learn from multi-modal information to capture modality-aware user–item interactions, providing a dynamic framework for modeling intricate behaviors and producing more nuanced embeddings.

CL, RL, AL APPLICATION

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LIMITATION & FUTURE DIRECTION

Limitation

1
- Current models using SSL techniques are limited by **evaluation settings focused on a single dataset**

2
- Effectively leveraging the **textual capabilities** of LLMs remains a key open challenge in recommendation research

3
- Assuming fixed user/item sets and the sparsity of new data **limit models' ability to adapt**, even with prompt tuning.

4
- SSL enhances recommender systems, but a solid theoretical foundation for various paradigms is still **needed to explain their benefits in recommendations**.

LIMITATION & FUTURE DIRECTION

Future Direction

1

- Recent research in CV and NLP has enhanced model generalization through SSL using CL or generative learning on large datasets.

2

- Diffusion models offer strong generative power for augmentation and inference, making them a promising future direction in recommendation.

3

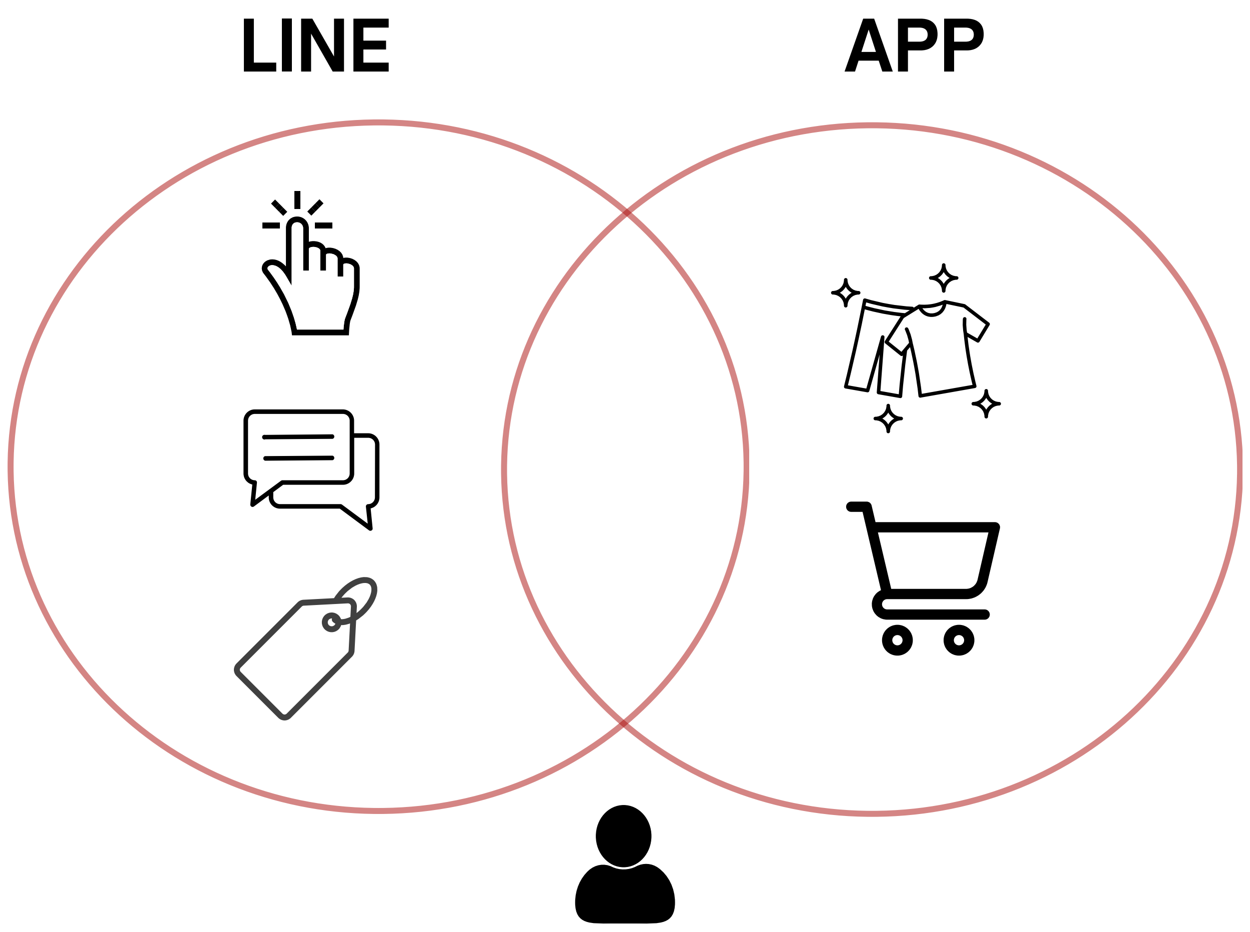
- SSL techniques are poised to play a crucial role in enhancing recommenders with the incorporation of LLMs.

4

- If recent paradigms can be created, it could reveal the principles and mechanisms behind SSL in recommendation systems, providing insights into the generalization and robustness of these algorithms.

BRIDGE GAP BETWEEN LINE to App with CroRec

Treat a user's cross-platform activity as different 'views' of the same person.



How it Works

Maximize the similarity between a user's LINE interactions and their App purchase history. By optimizing an InfoNCE objective, it learns that both views belong to the same person, creating a unified embedding.

Outcome

- **Solves Cold-Start Problem:** Provide relevant recommendations to new App users by leveraging their existing LINE history.
- **True O2O Personalization:** Create a single, "domain-invariant" customer profile that powers a seamless journey from online engagement to in-store purchase.

Data

- Source Domain: LINE behavioral data
- Target Domain: App data

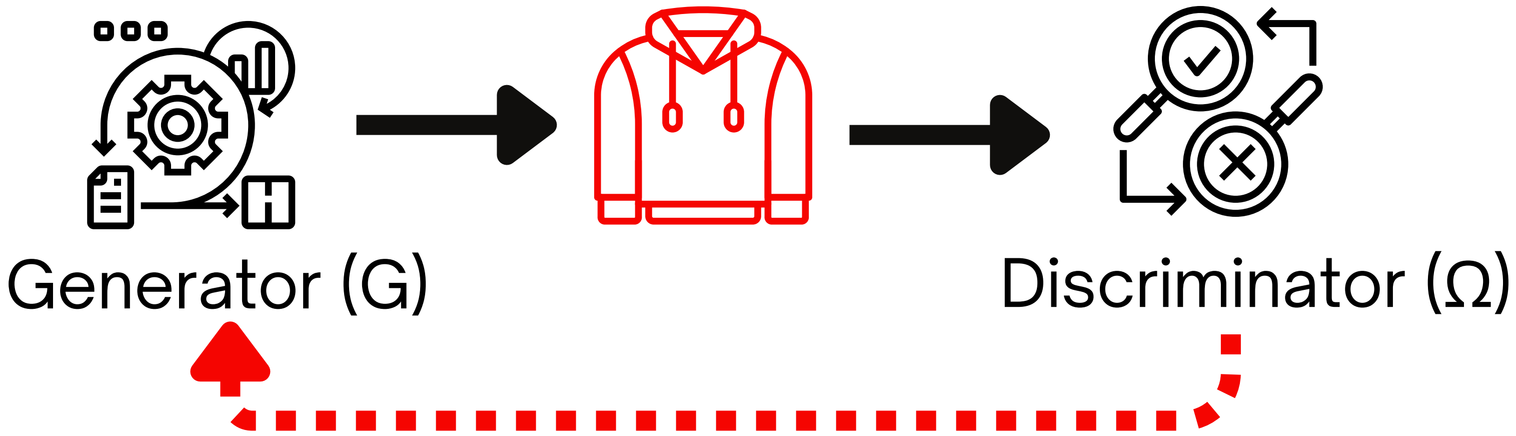
RL & AL - PREDICTING PURCHASES & ENHANCING QUALITY

Predicting with RL

Randomly "masking" an item in a customer's purchase history, the model learns to reconstruct the missing piece and explore the deep patterns behind a user's style.

Highly accurate predictions of "Next LifeWear Purchase". Allows proactive marketing campaigns to accurately grasp customer needs, thereby improving conversion rates and customer loyalty.

Improving Diversity with AL



A "Generator" creates recommendations, while a "Discriminator" tries to distinguish them from real purchases.

Breaks recommendation "bubbles" by surfacing compelling long-tail items (e.g.,a unique UT collaboration). Increases robustness against noisy data, ensuring recommendation quality and diversity.

SUMMARY - CL, RL, AL APPLICATION IN UQ

Technique	Solve the Problem	Unlock Value by
Constructive Learning(CL)	Disconnected LINE&APP data	Unify the customer view & solving cold-start issues
Reconstructive Learning(RL)	Predicting future purchases	Drive proactive, highly relevant recommendations
Adversarial Learning(AL)	Recommendation Stagnation & Noise	Increase diversity & ensure model robustness

APPENDIX

RL APPLICATION

General Collaborative Filtering

- **Reconstruction Target**
  - VAE and diffusion-based method: ratings to items
  - VAE-based: generate item content to achieve hybrid generation
  - graph-based method: valuable edges in the graph are masked and then reconstructed

- **Reconstructive Learning Paradigm**

VAE → MAE →denoised diffusion

Sequential Recommendation

- **Reconstruction Target**

1. Capture user behaviors and enhance recommendations
2. Enrich sparse data for model training
3. Reconstruct user–item interactions

- **Reconstructive Learning Paradigm**

SeqRec's RL mainly use mask-autoencoding method

Knowledge-Aware Recommendation

- **Reconstruction Target**

Denoise the KG by removing noisy and redundant triplets that are irrelevant to recommendation, and to generate valuable interactions within the KG.

- **Reconstructive Learning Paradigm**

KGRec include masked autoencoding and denoising diffusion approaches.

SCORE ->mask & restruct ->encoding

DiffKG employs the concept of denoising diffusion by adding Gaussian noise to the triplets in the KG and subsequently removing it.

RL APPLICATION

Cross-domain Recommendation

- **Reconstruction Target**

1. Serve as a fundamental mechanism to **bridge information gaps** between domains by reconstructing missing or incomplete data representations
2. **RA/SA-VAE** & **VDEA** adopt the user rating reconstruction paradigm, **DiffCDR** focuses on reconstructing user features

- **Reconstructive Learning Paradigm**

CroRec
VAE
RA/SA-VAE **VDEA** **DiffCDR**

Bundle Recommendation

autoencoder

- **Reconstruction Target**

Mask and reconstruct critical structures to enhance model robustness

- **Reconstructive Learning Paradigm**

Bundle Recommendation in RL include knowledge distillation and **masked autoencoding mechanism.**

RL w/ KD +MAE — Teacher GNN — MAE

Multi-Behavior Recommendation

- **Reconstruction Target**

Reconstruct behavior-specific patterns to enhance representation learning

- **Reconstructive Learning Paradigm**

Leverage variational autoencoding mechanisms to model behavior heterogeneity and capture complex interaction patterns

Multi-Behavior Recommendation **BVAE** **VCGAE**

RL APPLICATION

Multi-Modal Recommendation

- **Reconstruction Target**

Reconstruct graph structural information

- **Reconstructive Learning Paradigm**

integrates multi-modal information through VAE
features → **MVGAE** assigns μ&σ^2 →user-item graph node & employs product-of-experts →encoder→reconstruct

AL APPLICATION

General Collaborative Filtering

- IRGAN (diff)
- ABinCF & AugCF (non diff)

To improve recommendation indices, user ratings, or even perform data augmentation

Discrimination targets include sampled indices of items, the generated user purchase vector, or popular-niche item pairs for improved long-tail recommendation

Sequential Recommendation

- SAO, SRecGAN, SSRGAN (diff)
- AOS4Rec, MFGAN(non diff)

Distinguish between real & generated item sequences, optimizing generators to produce realistic recommendations

Discrimination targets are diverse, including encoded features derived from the item sequence , or the ranking score of items.

Social Recommendation

- RSGAN (diff)
- Adit et al., DASO (non diff)

Use AL to extract clean and representative social relations from the user-user graph in SocRec.

Effectively integrating social and collaborative information. To improve the generator's ability to produce accurate social-based recommendations

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AL APPLICATION

Cross-domain Recommendation

- RecGURU, DA-DAN (diff)

AL employs adversarial domain adaptation to generate domain-independent features, encoding domain-invariant user interaction preferences.

distinguishing domain-specific features to promote domain-invariant representation learning.

Multi-Modal Recommendation

- MMSSL (multi-modal adversarial mechanisms)

Perform structure learning based on multi-modal information, revealing modality-aware user-item interactions

Distinguishes real and generated user-item interactions. The user-item graph structure learned is split into user-specific interaction vectors.

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